Section	Details
Project	Generic Fresher Data Engineering + Analytics POC
Purpose	Freshers can use any self-created dataset and complete SQL, Azure Data Factory pipelines, KPI generation, and Power BI dashboard tasks
Applicable For	Manufacturing, IoT, Retail, Finance, HR, Healthcare, Logistics, Sales, Inventory, Customer, or any custom dataset
Main Goal	Build raw/staging/silver/gold tables, create ADF pipelines/data flows, write SQL tasks, generate KPIs, and build Power BI dashboards
Cloud / ETL Focus	Azure Data Factory, Azure SQL Database, Azure Data Lake Storage Gen2, Mapping Data Flows
Expected Deliverables	SQL scripts, ADF pipeline design, ADF data flow logic, KPI tables, Power BI dashboard, documentation
Important Instruction	Freshers should replace placeholder names such as <dataset_name>, <source_file>, and <business_key> based on their own dataset.

Layer	Suggested Table Name	Purpose	Example	Who Creates It
Raw / Bronze	raw_<dataset_name>	Store original ingested data with minimal/no transformation	raw_sensor_data / raw_sales_data	Fresher
Staging	stg_<dataset_name>	Clean data types, rename columns, handle nulls, remove duplicates	stg_sensor_data / stg_sales_data	Fresher
Silver	silver_<business_process>	Business-ready transformed data with derived columns	silver_machine_operations / silver_order_summary	Fresher
Gold	gold_<dataset_name>_kpi	Aggregated KPI tables used by dashboards	gold_sensor_kpi / gold_sales_kpi	Fresher
Anomaly / Exception	gold_<dataset_name>_exceptions	Stores invalid, outlier, failed, or suspicious records	gold_sensor_exceptions	Fresher
Audit	pipeline_run_audit	Track ADF pipeline status, start/end time, record counts, and errors	pipeline_run_audit	Fresher
Data Quality	dq_validation_results	Store null checks, duplicate checks, datatype checks, and business rule checks	dq_validation_results	Fresher

Column Name	Suggested Data Type	Description	Nullable	Business Meaning	Example
<primary_key_or_id>	INT / BIGINT / VARCHAR	Unique identifier if available	No	Unique row/business identifier	order_id, machine_id, employee_id
<event_datetime>	DATETIME	Date/time of business event	No	Used for trends and time-based analysis	order_date, reading_time
<category_column>	VARCHAR	Category/dimension field	Yes	Used for grouping and comparison	product_category, department, machine_type
<numeric_metric_1>	DECIMAL	Main measurable value	Yes	Used for KPI calculation	sales_amount, temperature, runtime
<numeric_metric_2>	DECIMAL	Secondary measurable value	Yes	Used for comparison or efficiency metrics	cost, target_value, quantity
<status_column>	VARCHAR / INT	Status indicator	Yes	Used for filtering and exception reporting	success/failure, active/inactive
ingestion_timestamp	DATETIME	Timestamp when data was loaded into raw layer	No	Audit column	ADF load time
source_file_name	VARCHAR	Name of uploaded source file	No	Traceability column	sales_jan.csv

Module	Task	Expected Output	Tools Suggested
Dataset Understanding	Understand business meaning of columns in their own dataset	Dataset summary and data dictionary	Excel, SQL
Raw Table Design	Design raw_<dataset_name> table using actual columns	Raw table DDL	Azure SQL / SQL Server
ADF Ingestion Pipeline	Create ADF pipeline to copy source data into raw table or data lake	Pipeline: PL_01_Raw_Ingestion	Azure Data Factory Copy Activity
ADF Mapping Data Flow	Create mapping data flow to clean and transform raw data	Data Flow: DF_01_Raw_To_Staging	ADF Mapping Data Flow
Staging Table Creation	Create stg_<dataset_name> with cleaned and standardized columns	Staging table	SQL, ADF Data Flow
Silver Transformation	Create business-level transformed table	silver_<business_process>	SQL, ADF Data Flow
Gold KPI Generation	Create aggregated KPI table for dashboard	gold_<dataset_name>_kpi	SQL, ADF Stored Procedure Activity
Data Quality Checks	Check nulls, duplicates, invalid values, and business rule failures	dq_validation_results	SQL, ADF Data Flow
Exception Handling	Separate invalid/outlier records into exception table	gold_<dataset_name>_exceptions	SQL, ADF Conditional Split
Pipeline Orchestration	Connect activities in sequence with success/failure paths	End-to-end ADF workflow	ADF Pipeline Activities
Power BI Dashboard	Build dashboard using silver/gold tables	Interactive dashboard	Power BI
Documentation	Document data flow, assumptions, table names, and screenshots	Final POC document	Word / PPT

Task ID	SQL Requirement	Input Table	Expected Output	Concept
SQL-1	Create raw_<dataset_name> table based on actual dataset columns	Source file schema	Raw table DDL	CREATE TABLE
SQL-2	Create stg_<dataset_name> with cleaned column names and correct data types	raw_<dataset_name>	Staging table DDL	DDL, CAST/CONVERT
SQL-3	Find duplicate records using business key or all columns	raw_<dataset_name>	Duplicate records report	GROUP BY, HAVING
SQL-4	Find null count and null percentage for important columns	raw/stg table	Data quality summary	COUNT, CASE WHEN
SQL-5	Convert date/time columns into standard format	raw_<dataset_name>	Cleaned datetime columns	CONVERT, DATE functions
SQL-6	Create derived KPI columns based on dataset context	stg_<dataset_name>	silver_<business_process>	CASE WHEN, calculated fields
SQL-7	Create daily/weekly/monthly KPI summary	silver_<business_process>	gold_<dataset_name>_kpi	GROUP BY, aggregation
SQL-8	Identify anomalies/outliers using business rules	silver_<business_process>	Exception records	CASE WHEN, filters
SQL-9	Create rank/leaderboard reports	gold_<dataset_name>_kpi	Ranked output	RANK, DENSE_RANK
SQL-10	Create incremental load logic using date/key column	raw/stg table	MERGE-ready logic	MERGE, UPSERT
SQL-11	Create audit table for pipeline runs	ADF pipeline metadata	pipeline_run_audit	INSERT audit rows
SQL-12	Create Power BI reporting views	silver/gold tables	vw_<dataset_name>_dashboard	CREATE VIEW

Pipeline Name	Pipeline Type	Source	ADF Activities	Target	Purpose	Expected Output
PL_01_Raw_Ingestion	Batch Ingestion	CSV / JSON / Excel / SQL source	Get Metadata → Copy Data → Stored Procedure for audit	raw_<dataset_name> or ADLS Bronze folder	Load source data as-is	Raw data available for processing
PL_02_Raw_To_Staging	Cleaning Pipeline	raw_<dataset_name>	Mapping Data Flow → Derived Column → Select → Sink	stg_<dataset_name>	Clean datatypes, rename columns, remove duplicates	Clean staging table
PL_03_Data_Quality_Checks	Validation Pipeline	raw/stg_<dataset_name>	Lookup → If Condition → Stored Procedure	dq_validation_results	Validate nulls, duplicates, invalid values	DQ pass/fail report
PL_04_Exception_Handling	Exception Pipeline	stg_<dataset_name>	Mapping Data Flow → Conditional Split → Sink	gold_<dataset_name>_exceptions	Separate invalid or outlier records	Exception table
PL_05_Staging_To_Silver	Business Transformation Pipeline	stg_<dataset_name>	Mapping Data Flow → Join/Aggregate/Derived Column → Sink	silver_<business_process>	Create business-ready table	Silver analytics table
PL_06_KPI_Gold_Load	Aggregation Pipeline	silver_<business_process>	Stored Procedure / Data Flow Aggregate → Sink	gold_<dataset_name>_kpi	Generate KPI summary	Gold KPI table
PL_07_Dashboard_Refresh	Reporting Pipeline	gold_<dataset_name>_kpi	Web Activity / Power BI Refresh API / Manual refresh	Power BI dataset	Refresh dashboard after data load	Updated Power BI dashboard
PL_08_Incremental_Load	Incremental Pipeline	Source with date/key column	Lookup watermark → Copy Data → Stored Procedure update watermark	raw/stg tables	Load only new or changed records	Incremental data load
PL_09_Master_Orchestration	Orchestration Pipeline	All child pipelines	Execute Pipeline activities with success/failure dependency paths	All target layers	Run end-to-end workflow	Complete automated POC pipeline
PL_10_Pipeline_Monitoring	Audit Pipeline	ADF system variables	Set Variable → Stored Procedure → Web/Email alert	pipeline_run_audit	Track pipeline run status and failures	Audit and failure logs

Step	ADF Component	What Happens	Example for Any Dataset	Output
1	Linked Service	Create connection to source and target systems	Connect to ADLS, Azure SQL, Blob Storage, or on-prem SQL	Reusable connection
2	Dataset	Define source and target file/table structures	CSV dataset, JSON dataset, Azure SQL table dataset	Dataset objects
3	Pipeline	Create workflow that controls activities	PL_01_Raw_Ingestion	Pipeline canvas
4	Get Metadata Activity	Check whether file exists and capture file details	Validate source file path and file name	Metadata output
5	Copy Data Activity	Move data from source to raw table/folder	Copy CSV/JSON/Excel to raw_<dataset_name>	Raw layer loaded
6	Mapping Data Flow - Source	Read raw table/file as input inside data flow	Source = raw_<dataset_name>	Raw stream
7	Select Transformation	Rename columns and keep required columns	Rename CustomerName to customer_name	Standardized columns
8	Derived Column Transformation	Create calculated fields and convert data types	Convert amount to decimal, create order_month	Clean derived columns
9	Filter Transformation	Remove invalid or unwanted rows	Remove records with null business key	Filtered records
10	Conditional Split	Split valid and invalid records	Valid rows to staging, invalid rows to exception table	Valid and exception streams
11	Aggregate Transformation	Create KPI-level summaries	Total sales by month, avg temperature by machine	KPI dataset
12	Sink Transformation	Write transformed data into target table/folder	Sink = stg_<dataset_name> / gold_<dataset_name>_kpi	Loaded target
13	Stored Procedure Activity	Run SQL logic after data movement	Update audit table, update watermark, refresh KPI table	Post-load processing
14	If Condition Activity	Branch pipeline based on DQ result or row count	If failed_count > 0 then mark pipeline failed	Controlled execution
15	Execute Pipeline Activity	Call child pipelines from master pipeline	Master pipeline calls ingestion, staging, KPI, dashboard refresh	End-to-end orchestration
16	Monitoring	Track pipeline run status, duration, and errors	ADF Monitor tab + pipeline_run_audit table	Operational visibility

Flow Order	Layer / Component	Generic Data Flow
1	Source	Self-created dataset: CSV / JSON / Excel / SQL table / API extract
2	ADF Linked Service	Connection to source and target
3	ADF Dataset	Metadata definition for source and sink
4	ADF Copy Activity	Move source data into Raw/Bronze layer
5	Raw Table	raw_<dataset_name> stores original data
6	ADF Mapping Data Flow	Clean, cast, rename, filter, deduplicate
7	Staging Table	stg_<dataset_name> stores cleaned data
8	ADF Mapping Data Flow / SQL	Apply business transformations and derived columns
9	Silver Table	silver_<business_process> stores analysis-ready data
10	SQL Stored Procedure / Data Flow Aggregate	Create KPI summaries
11	Gold Table	gold_<dataset_name>_kpi stores dashboard-ready metrics
12	Power BI	Build dashboard and visuals using gold/silver tables
13	ADF Audit	pipeline_run_audit captures execution status and row counts
14	ADF Monitor	Monitor success, failure, duration, and errors

Dashboard Requirement	Visual Type	Input Layer	Purpose
KPI Overview	Cards	gold_<dataset_name>_kpi	Show main metrics for the dataset
Trend Analysis	Line Chart	silver_<business_process>	Analyze changes over time
Category Comparison	Bar / Column Chart	silver/gold tables	Compare metrics across categories
Detail Table	Table / Matrix	silver_<business_process>	Show filtered detailed records
Anomaly Monitoring	Table / Matrix with conditional formatting	gold_<dataset_name>_exceptions	Highlight abnormal or failed records
Data Quality Summary	Cards + table	dq_validation_results	Show data quality pass/fail metrics
Pipeline Monitoring	Cards + table	pipeline_run_audit	Track ADF pipeline executions
Interactive Filters	Slicers	Date/category/status columns	Allow users to explore the dataset
Drill-through Page	Drill-through report	silver table	Show detailed records behind each KPI

Deliverable	Description
Dataset Summary	Explain the dataset they created, columns, business problem, and assumptions
Data Model	Raw, staging, silver, gold, DQ, and audit table names
SQL Scripts	DDL, transformation, KPI, DQ, anomaly, and reporting queries
ADF Pipelines	Ingestion, raw-to-staging, staging-to-silver, KPI load, DQ, audit, and master orchestration pipelines
ADF Data Flow Screenshots	Screenshots of source, select, derived column, filter, conditional split, aggregate, and sink transformations
Power BI Dashboard	Dashboard using silver/gold tables
Final Documentation	Architecture, pipeline flow, SQL logic, dashboard screenshots, and learnings

Area	Weightage	Evaluation Criteria
Dataset Understanding	10%	Clear explanation of dataset and business use case
SQL Skills	20%	Correct table design, queries, KPIs, DQ checks, and optimization
ADF Pipeline Design	25%	Correct use of Copy Activity, Mapping Data Flow, orchestration, parameters, and audit
Data Modelling	15%	Proper raw/staging/silver/gold structure
Power BI Dashboard	15%	Useful visuals, slicers, KPI cards, and business insights
Documentation	10%	Clear screenshots, flow explanation, assumptions, and final summary
Innovation	5%	Incremental load, alerts, streaming simulation, ML/anomaly logic, reusable parameters

Pipeline Type	Scenario	Expected Outcome
Batch Ingestion	Load CSV/JSON/Excel into raw tables	Raw data ingestion
Streaming Pipeline	Process near real-time records	Streaming analytics
ETL Pipeline	Transform raw to staging/final	Business-ready data
Data Quality Pipeline	Validate nulls, duplicates, invalid values	DQ reports
KPI Pipeline	Generate daily/monthly KPIs	KPI summary tables
Dashboard Refresh Pipeline	Refresh reporting datasets	Updated dashboard
Monitoring Pipeline	Track pipeline status and failures	Audit reporting